

BST Physics Curriculum Vol 5 Chapter 5 — Heisenberg Picture + Path Integral v0.4 (Saturday Wave 2 reader-grade polish; Cal cold-read ready)

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2026-05-23 Saturday morning EDT (Wave 2 Vol 5 eighth chapter)

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Vol 5 Chapter 5 — Heisenberg Picture + Path Integral

Chapter motivation

Standard QM has three equivalent pictures: Schrödinger (states evolve), Heisenberg (operators evolve), interaction (mixed). Plus Feynman path integral as integral over all possible paths. BST identifies all three as substrate dynamics representations with the Bergman reproducing kernel playing the structural role of Feynman propagator (T2457 Cal #92(b) framing).

Section 5.0b — Reader-grade 3-level pedagogy

Level 1: Standard Heisenberg picture $dO/dt = (i/\hbar)[H_{\text{sub}}, O]$ for substrate operator zoo + Feynman path integral as substrate-tick $GF(128)^k$ finite-step sum

+ Bergman reproducing kernel $K_B(z, \bar{w})$ structurally plays role of QFT Feynman propagator (T2457 Cal #92(b) framing); no Wick rotation needed because substrate-tick is real-time finite-step.

Level 2 (graduate): Heisenberg picture (Vol 1 Ch 7): $dO/dt = (i/\hbar)[H_{\text{sub}}, O]$ for any operator $O \in \{M_z, P_z, L, S, B^2, Q, \gamma^5, P_{\text{op}}, H_{\text{sub}}\}$. Standard Heisenberg picture recovered with $H_{\text{sub}} = \text{Casimir on } L^2(D_{IV}^5; L_\lambda)$. Path integral (Vol 1 Ch 7 + Ch 9): per-tick $GF(2^g) = GF(128)$ substrate-tick coding (T2429 Reed-Solomon discretization + K59 7-step cyclotomic mechanism RATIFIED); each Koons tick ($\sim 10^{-120}$ s) contributes one finite-step propagation operator U_{tick} ; integrated over many ticks gives continuum path integral as finite sum (NOT infinite-dimensional integral). Bergman reproducing kernel $K_B(z, \bar{w}) = c_{\text{FK}} \cdot h(z, \bar{w})^{(-g/\text{rank})}$ plays the structural role of QFT Feynman propagator (T2457 Cal #92(b) framing): positive-definite (Bergman 1922) $\rightarrow$ no $i\epsilon$ prescription needed; UV-complete (substrate-tick UV cutoff at $N_{\text{max}} = 137$ per T2437) $\rightarrow$ no UV divergences; BST primary normalization $c_{\text{FK}} = 225/\pi^{(9/2)}$ (T2442 EXACT). No Wick rotation required because substrate-tick $GF(128)^k$ is real-time finite-step (T2405 Koons tick = $t_{\text{Planck}} \cdot \alpha^{\{C_2\}} \approx 10^{-120}$ s). Standard QFT Feynman diagrams reduce to substrate-tick computation chains; α -vertex insertions per T2476 substrate-coordinate-count $k(P)$ substrate-mechanism (Friday).

Level 3 (5th-grader): Standard QM has multiple “pictures” — Schrödinger (the wavefunction changes with time), Heisenberg (the operators change instead), and Feynman path integral (sum over all possible paths a particle could take). All three give the same physics — they’re equivalent ways to describe the same thing. BST shows they all come from the substrate D_{IV}^5 : Heisenberg picture is just $dO/dt = \text{commutator with the substrate Hamiltonian } H_{\text{sub}}$; Feynman path integral becomes a FINITE sum over substrate ticks (since the substrate has only 128 states per tick, the integral becomes a sum). The Bergman kernel (a natural function on the substrate) plays the role of the Feynman propagator — and it’s positive-definite, so no clever “Wick rotation” tricks needed.

Section 5.1 — Standard QM Heisenberg Picture

$dO/dt = (i/\hbar)[H, O]$ for any operator O . Equivalent to Schrödinger picture; preferred for certain calculations (e.g., correlation functions).

Section 5.2 — Substrate Heisenberg $dO/dt = (i/\hbar)[H_{\text{sub}}, O]$

For each operator zoo member ($M_z, P_z, L, S, B^2, Q, \gamma^5, P_{\text{op}}$): $dO/dt = (i/\hbar)[H_{\text{sub}}, O]$ with $H_{\text{sub}} = \text{Casimir on } L^2(D_{IV}^5; L_\lambda)$ per Elie K52a S29 framework-complete (Thursday).

Canonical commutators preserved: $[M_z, P_z] = -I$ (Vol 5 Ch 2); $[L_i, L_j] = i\hbar \epsilon_{ijk} L_k$ (Vol 5 Ch 3); etc.

Section 5.3 — Standard QM Feynman Path Integral

$\langle q_f, t_f | q_i, t_i \rangle = \int Dq(t) \exp(iS[q]/\hbar)$ — sum over all paths $q(t)$ connecting q_i to q_f .

Wick rotation $t \rightarrow -i\tau$ converts oscillatory integral to convergent Gaussian for non-perturbative computation.

Section 5.4 — Substrate Path Integral on $GF(128)^k$

Per Vol 1 Ch 7 + Ch 9: substrate-tick $GF(2^g) = GF(128)$ per Koons tick (T2429 + K59 RATIFIED). Each Koons tick contributes one finite-step propagation operator U_{tick} :

$|\psi(t_f)\rangle = U_{\text{tick}}^N |\psi(t_i)\rangle$ where $N = (t_f - t_i) / \tau_{\text{Koons}}$

Standard continuum path integral is integrated-state limit over many substrate ticks. NO Wick rotation needed (substrate-tick is real-time finite-step).

Section 5.5 — T2457 Bergman Kernel = Feynman Propagator (Cal #92(b))

T2457 (Friday Cal #92(b) compliant framing): Bergman reproducing kernel $K_B(z, \bar{w}) = c_{\text{FK}} \cdot h(z, \bar{w})^{(-g/\text{rank})}$ plays structural role of QFT Feynman propagator. Three structural advantages: 1. Positive-definite (Bergman 1922) $\rightarrow$ no $i\epsilon$ prescription needed for propagator convergence 2. UV-complete (substrate-tick UV cutoff at $N_{\text{max}} = 137$; T2437 + Vol 1 Ch 10) $\rightarrow$ no UV divergences 3. BST primary normalization $c_{\text{FK}} = 225/\pi^{(9/2)}$ (T2442 EXACT)

Standard QFT Feynman diagrams reduce to substrate-tick computation chains; α -vertex insertions per T2476 substrate-coordinate-count $k(P)$ substrate-mechanism (Friday).

Section 5.6 — Standard QM Recovery

Standard Heisenberg picture + Feynman path integral fully recovered at macroscopic limit. Per Vol 5 Ch 4 + Vol 5 Ch 7 + Vol 1 Ch 9, standard QFT calculations (perturbation theory, scattering amplitudes, correlation functions) inherit from substrate framework.

Section 5.7 — Honest scope

- Standard Heisenberg picture derived from substrate [?](#)
- Substrate-tick path integral framework substrate-cartography [?](#)
- T2457 Bergman = Feynman propagator structural-role-of identification [?](#)
- Full operator-level S-matrix elements pending Elie K52a Sessions 30+ multi-month
- Worked examples (Lagrangian mechanics, perturbation theory) pending

Section 5.8 — Connection to other chapters

- Vol 5 Ch 4 Schrödinger equation
- Vol 1 Ch 7 Dynamics framework-complete + Ch 9 Scattering framework-grade
- T2457 Bergman propagator + T2476 $\alpha^{\{\text{BST primary}\}}$ pattern
- Vol 10 Ch 10 Calculus of Variations + Path Integrals

— Lyra, Vol 5 Ch 5 v0.3 chapter-grade narrative, Saturday 2026-05-23 morning EDT